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Innate immune antimicrobial response by retinoic acid during reactive adipogenesis is dependent on HIF1α

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Cutaneous bacterial infection by Gram-positive bacteria induces an essential innate immune response in dermal fibroblasts resulting in rapid differentiation of a subset of fibroblasts into adipocytes (reactive adipogenesis). Induction of adipogenesis is accompanied by a large yet transient increase in production of the antimicrobial peptide (AMP) cathelicidin (Camp) by preadipocytes during transition to mature adipocytes. Exposure to retinoic acid (RA) further enhanced *Camp* expression in 3T3-L1 preadipocytes (129-fold vs. control $p < 0.0001$) and in vivo in mouse skin (7-fold vs. control, $p < 0.001$). When RA was present during bacterial infection with *S. aureus*, *Camp* expression increased 19-fold, $p < 0.001$. To understand the mechanism by which RA regulates Camp we examined the role of Hypoxia-inducible factor-1α (HIF-1α). Previous studies have shown that retinoic RA enhances *hif1a* expression in several cell types, but a connection between RA, HIF-1α and AMPs had not previously been suspected. In concordance with our hypothesis, a HIF-1α transcriptional binding site is present in the Camp promoter, and RA-treated 3T3-L1 cells exhibited ∼2-fold higher *hif1a* mRNA expression compared to differentiated controls ($p < 0.001$). Immunostaining and Western blot also showed significantly higher levels of HIF-1α after RA treatment in vitro. In mice, RA treatment resulted in nearly a 2-fold increase in *hif1a* expression ($p < 0.001$), while a ∼3-fold, $p < 0.0001$ increase was seen in mice treated with RA during S. aureus infection. Pharmacological induction of HIF-1α by addition of the agonist (mimosine) in 3T3-L1 cells resulted in enhanced *Camp* expression comparable to RA (37- and 30- fold respectively vs. control, $p < 0.0001$) while cotreatment of RA and mimosine was synergistic (112-fold vs. control, $p < 0.0001$). HIF-1α inhibition by the inhibitor acriflavine abolished *Camp* expression following RA treatment. These findings link HIF-1α and RA in the capacity of dermal adipogenesis to fight infection.

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A novel facial treatment cleansing device improves acne skin

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Acne vulgaris is a common skin disease that impacts an estimated 50 million people in the US annually. Acne often affects teenagers and young adults, however in more recent years women in their 30s and 40s are increasingly affected. There are several over-the-counter (OTC) drugs available, including actives such as benzoyl peroxide and salicylic acid. In order to improve the efficacy of acne actives, different strategies are created, such as using newly approved OTC adapalene and/or dermatologist prescribed drugs including antibiotics. Gentle cleansing with removal of excess sebum and skin debris is an important first step in any acne treatment. Recently, we developed and launched a novel mechanical facial treatment cleansing device with accompanying treatment cleansers, that can exfoliate the skin by gently stimulating desquamation and remove deposits on the face, maintaining and improving skin health. An extra gentle cleansing surface was developed to support subjects with sensitive skin. In two separate clinical studies, we evaluated different surfaces with an acne treatment cleanser containing 0.5% salicylic acid and assessed the efficacy of improvement when the mechanical treatment cleansing device was used. In one study, 40 subjects, ages 16-25, were recruited and in the second study, 30 subjects, ages 18-45, were recruited. Subjects were followed for 12 weeks to evaluate skin health. In the first study, subjects used the device alone randomized to half of their face, while using the acne treatment cleanser on both sides. Subject's Investigator Global Acne Assessment (IGA) improved throughout the 12-week period in both studies. In the first study, both sides of faces showed improvement. However, the device side improved faster than the treatment cleanser alone but by Week 12, the difference was no longer statistically significant. In the second study, additional skin attributes such as softness, smoothness, texture and clarity showed improvements. These studies demonstrated that the use of proper skin treatment cleansing tools can enhance acne treatment.

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Integrated stress response facilitates cutaneous wound healing

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A key component in the resolution of cutaneous wounds is the re-epithelialization of the wound bed by migrating keratinocytes using a mechanism called collective cell migration (CCM). Resolution of wounds induces stress in the skin and a central mechanism used by cells to sense and respond to stress signals is the Integrated Stress Response (ISR). Distinct stress-activated protein kinases phosphorylate the translation factor eIF2 (eIF2-P), leading to diminished global protein synthesis coincident with preferential translation of stress-related gene transcripts. Previously, we showed that the ISR is critical for keratinocyte viability in response a range of environmental stresses. Therefore, we hypothesized that the ISR plays a central role in wound healing. Wounding of immortalized NTERT human keratinocyte monolayers led to the rapid activation of the eIF2 kinase GCN2, and subsequent eIF2-P and translational control. To address the function of GCN2 in wound healing, we generated NTERTs lacking GCN2 using a CRISPR/CAS9 system. Deletion of GCN2 in wounding assays led to a loss of eIF2-P and restoration of translation, indicating that GCN2 is the eIF2 kinase activated during wound healing. In a high-throughput wound healing assay, deletion of GCN2 significantly impaired keratinocyte CCM compared to wild-type keratinocytes. Treatment of GCN2-deficient cells with guanabenz, a pharmaceutical agent that thwarts the dephosphorylation of eIF2 and enhances the ISR, rescues CCM. These results indicate that GCN2 and the ISR are required for the induction of CCM during wound healing. We have identified genes involved in wound healing whose expression are regulated by GCN2 and we are currently using genome-wide assays for translational control to determine the pathways regulated by GCN2. In conclusion, GCN2 is critical for re-epithelialization of cutaneous wounds and we propose the ISR is a potential therapeutic target in chronic non-healing wounds.

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Smad Anchor for Receptor Activation (SARA) prevents skin fibrosis via inhibiting pericyte transdifferentiation to myofibroblasts

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Myofibroblast activation is a major feature of skin fibrosis, but the origins of myofibroblasts are yet uncertain. While pericytes have been suggested as myofibroblast progenitors in various organs, their contribution to skin fibrosis is not clear. Our previous data suggest that SARA is an anti-fibrotic molecule that acts by modulating cell phenotype. Here, we studied a role for SARA in pericyte activation in a mouse model of skin fibrosis. To test role of SARA, we generated a mouse that constitutively expresses human full-length SARA cDNA specifically in pericytes under the control of PDGFRβ promoter-driven Cre recombinase activity ($SARA^{Tg}$; PDGFRβ-Cre), and further crossed them with the Z/EG strain to mark the PDGFRβ+ pericytes with GFP. Thickening of the dermal layer and $PDGFR\beta^{+}GFP^{+}$ cell expansion in dermis were significant in $SARA^{Tg-}$ mice treated with bleomycin compared to healthy controls, whereas the changes were less overt in $SARA^{Tg+}$; Z/EG; PDGFRβ-Cre mice (mean dermal thickness; 166 and 156 μm in $SARA^{Tg-}$ or $^{Tg+}$ mouse without bleomycin, 222 and 147 um in $SARA^{Tg-}$ or $^{Tg+}$ mouse with bleomycin). The majority of control $PDGFR\beta^{+}GFP^{+}$ cells were αSMA^{-}, while 56% of them were αSMA^{+} in skin subjected to bleomycin, suggesting that $PDGFR\beta^{+}$ cells transdifferentiated into myofibroblasts. Approximately 34% of $PDGFR\beta^{+}GFP^{+}$ cells were $SARA^{+}$ in healthy mouse skin, which was decreased to 12% in mice subjected to bleomycin. In contrast, SARA levels were preserved in $SARA^{Tg+}$ mice even after bleomycin injection. The reduction was also confirmed by SARA mRNA levels (1.09 vs. 0.59 relative expression, $P=0.018$). These data suggest that $PDGFR\beta^{+}$ pericytes contribute to skin fibrosis in mice, and SARA overexpression in pericytes prevents their transdifferention, hence preventing fibrosis. Therefore, SARA could be a novel therapeutic target in treating skin fibrosis.

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Defining the β-catenin transcriptional program in non-healing diabetic foot ulcers: new pathway to targeted therapy

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A growing body of evidence suggests that β-catenin, the major downstream driver of canonical Wnt signaling, is activated during physiologic wound healing and that aberrantly sustained activation may conversely inhibit wound healing. We have previously shown that nuclear localization of β-catenin in the epidermal edge of chronic ulcers is predictive of a non-healing clinical outcome. The purpose of this study was to identify the transcriptional targets of β-catenin signaling in wound edge biopsies from chronic diabetic foot ulcers (DFUs) using chromatin immunoprecipitation with high-throughput sequencing (ChIP-seq). Chromatin was extracted from $n=7$ FFPE non-healing DFUs (pre-screened for β-catenin nuclear localization); and following DNA sonication, fragments of the appropriate size were obtained for ChIP with β-catenin antibodies. Sequencing reads were mapped to the human genome and peak calling was performed using MACS. Ingenuity Pathway Analysis of β-catenin target genes revealed enrichment of multiple pathways including cytoskeletal organization (468 genes; $P=1.19\times10^{-24}$). In particular, a series of Rho family proteins, which regulate the assembly and organization of the actin cytoskeleton, were noted to be targets of β-catenin binding in the non-healing DFUs, including RhoA member genes ArhGEF3, ArhGAP12 and ArhGAP35. Western blotting and immunofluorescence analysis confirmed deregulation of these members in DFUs when compared to normal plantar foot skin. These findings support a transcriptional role for β-catenin in impairing keratinocyte migration and re-epithelialization in non-healing DFUs. Ultimately, this technique will help determine novel β-catenin-regulated processes contributing to impairment of healing in chronic ulcers.

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In search of the common mechano-chemical pathways during the regeneration of spiny (*acomys cahirinus*) and laboratory (*mus musculus*) mouse skin

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Following large full-thickness skin wounding, African spiny mice (*acomys cahirinus*) demonstrated robust regenerative ability, while laboratory mice (*mus musculus*) regenerate some hairs from the wound center. Although some studies have reported the molecular expression differences between the two species after wounding, whether there is a shared mechano-chemical mechanism mediating their common and different responses remains to be explored. We first characterized skin features of both laboratory and spiny mice such as hair types, hair cycling, and the response to small and large wounds. Spiny hair bulge is larger and shows high expression of stem cell markers, K15 and CD34. Spiny mouse skin contains a large portion of adipose tissue, which may contribute to its distinctively soft skin. RNA-seq analysis show that epithelial-to-mesenchymal transition, cell proliferation, and matrix remodeling-related genes were differentially upregulated in the wounds of both laboratory and spiny mice on post-wound day 15. Spiny mice showed high collagen III to I expression ratio in the wound, which is similar to that of fetal laboratory mice skin. We measured the spatiotemporal stiffness during wound healing of both mouse species and compared them to their hair regeneration pattern, and determined that there is an optimal range of tissue stiffness at 5-15 kPa for wound-induced hair neogenesis to occur. By manipulating the upstream transcription factor that initiates both the differential gene expression (chemical) and matrix-remodeling (mechanical) events, we were able to change the outcome of hair regeneration after wounding. These findings shed new lights on the regenerative biology of the skin in different mice, different types of WIHN, and may help us understand the fundamental mechanism of skin repair versus regeneration.